

A PROJECT REPORT

On

Driver Drowsiness Detection System

For the partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY

In

Computer Science and Engineering

Submitted By

Harsh Saini (2101920100124)

Harshit Nirmal (2101920100127)

Under the supervision of

Mr. Shivam Tiwari



G.L. Bajaj Institute Of Technology & Management

Greater Noida

Affiliated to



Dr. APJ Abdul Kalam Technical University

Lucknow

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Declaration

We hereby declare that the project work presented in this report entitled “**Driver Drowsiness Detection System**” in partial fulfilment of the requirement for the award of the degree of Bachelor of Technology in Computer Science & Engineering, submitted to A.P.J Abdul Kalam Technical University, Lucknow, is Based on my own work carried out at Department of Computer Science & Engineering G.L. Bajaj Institute of Technology & Management, Greater Noida. The work contained in the report is original and project work reported in this report has not been submitted by me/us for award of any degree or diploma.

Signature:

Name: Harsh Saini

Roll No:2101920100124

Signature:

Name: Harshit Nirmal

Roll No:2101920100127

Date:

Place: Greater Noida

Certificate

This is to certify that the project report entitled “**Driver Drowsiness Detection System**” done by **Harsh Saini (2101920100124)** and **Harshit Nirmal (2101920100127)** is an original work carried out by them in Department of Computer Science & Engineering G.L. Bajaj Institute of Technology & Management, Greater Noida under my guidance. The matter embodied in this project work has not been submitted earlier for the award of any degree or diploma to the best of my knowledge and belief.

Date:

Mr. Shivam Tiwari

Signature of the Supervisor

Dr.Sansar Singh Chauhan

Head of Department

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Abstract

A Driver Drowsiness Detection System is an important technology designed to improve road safety and reduce the number of accidents caused by drowsy driving. Drowsy driving is a serious issue that affects millions of people worldwide, and it can have deadly consequences. Fatigue can impair a driver's judgment, reaction time, and decision-making abilities, leading to serious accidents and even fatalities. To address this issue, the Driver Drowsiness Detection System has been developed.

This system works by continuously monitoring the driver's behavior and physiological markers, such as eye movement and head position, to detect any signs of drowsiness. The system uses cameras, sensors, or other devices to collect data, which is then analyzed in real-time to determine if the driver is becoming drowsy. If the system determines that the driver is indeed tired, it sounds an alarm or vibrates the seat to alert them and prevent them from falling asleep at the wheel. The aim of this technology is to prevent accidents and save lives by keeping drivers alert and focused on the road.

The Driver Drowsiness Detection System is easy to use and can be integrated into most modern vehicles. It operates automatically, without the need for any manual intervention, so drivers can simply concentrate on driving. The system is also highly accurate, with a low false-positive rate, meaning that it will only alert drivers if they are actually drowsy and in need of a break.

In conclusion, the Driver Drowsiness Detection System is a critical technology that can save lives and improve road safety. With the number of accidents caused by drowsy driving on the rise, it is more important than ever to adopt this technology and keep our roads safe for everyone. Whether you're a professional truck driver or a weekend road tripper, the Driver Drowsiness Detection System is an essential tool for ensuring that you stay awake, alert, and focused on the road at all times.

Table of Content

Declaration.....	(ii)
Certificate.....	(iii)
Acknowledgement.....	(iv)
Abstract.....	(v)
Chapter 1. Purpose of the Study.....	8
Chapter 2. Background of the Study.....	9
Chapter 3. Factors promoting to undertake the project.....	10
Chapter 4. Literature survey.....	11
Chapter 5. Aims and Objectives of Study.....	19
Chapter 6. Research Gaps.....	20
Chapter 7. Scope of Study.....	21
Chapter 8. Methodology.....	22
8.1 Flowchart of Model	22
8.2 Description of tools used	24
8.3 Use Case.....	25
8.4 Data Flow Diagram.....	25
8.5 Entity Relationship Diagram.....	26
8.6 Code.....	27
Chapter 9. Result and Discussion.....	28
Chapter 10. Conclusion.....	29
References.....	30

List of Figures

Figure 8.1.1	Flow Chart f Drowsiness Detection System.....	22
Figure 8.1.2	Facial Recognition Associated With Eyes.....	23
Figure 8.1.3	Facial Recognition Associated With Mouth.....	24
Figure 8.4.1	Data Flow Diagram.....	25
Figure 8.6.1	Implementation of Code.....	27
Figure 9.1	Result of Eyes & Mouth Aspect Ratio.....	30

Chapter-1

Purpose of study

The purpose of a study on a driver drowsiness detection system is to investigate the potential benefits and drawbacks of implementing such technology in reducing the risks associated with driving while fatigued. The study seeks to answer a number of key questions including:

- (i)** What physiological and behavioral signs indicate drowsiness in drivers and how can these signs be monitored and analyzed accurately?
- (ii)** Can current camera and machine learning technology accurately detect drowsiness in drivers?
- (iii)** What is the most effective way to alert a driver when drowsiness is detected?
- (iv)** How do drivers react to and feel about the use of drowsiness detection systems in their vehicles?
- (v)** What are the limitations and challenges facing current drowsiness detection systems and how can they be overcome?
- (vi)** What is the impact of drowsiness detection systems on road safety and can they effectively reduce the number of accidents caused by drowsy driving?

The results of the study can provide important information for the development and implementation of driver drowsiness detection systems and ultimately help to make our roads safer.

Chapter-2

Background of study

The roots of the investigation into driver fatigue detection systems can be traced back to the recognition of the fact that driving while drowsy is a major contributor to road accidents, leading to the need for a technological solution to this problem. The evolution of these systems has been propelled by advancements in technology such as the proliferation of low-cost sensors and the expansion of machine learning and computer vision. The examination of these systems encompasses a broad range of fields, including psychology, human-computer interaction, computer vision, and machine learning. The ultimate objective of the research is to create systems that are efficient in recognizing drowsiness and warning drivers promptly, thereby enhancing road safety and reducing the amount of accidents caused by driver fatigue.

Factors Promoting to Undertake the Project

- (i) **Enhanced safety:** The top reason for implementing driver drowsiness detection systems is to increase road safety. By alerting drivers when they are feeling fatigued, these systems can prevent accidents caused by drowsiness.
- (ii) **Regulatory compliance:** In many countries, there are regulations that mandate the installation of drowsiness detection systems in commercial vehicles like buses, trucks, and other vehicles carrying passengers or goods.
- (iii) **Increased efficiency:** By reducing the number of accidents caused by drowsy drivers, drowsiness detection systems can boost the efficiency of commercial vehicle operations. This can result in lower downtime and repair costs, leading to improved productivity.
- (iv) **Reduced risk:** Companies operating commercial vehicles can lower their risk by installing drowsiness detection systems. This includes reducing the probability of accidents caused by drowsy driving, as well as those resulting from other factors such as distracted driving.
- (v) **Technological advancements:** The continued development of new and advanced technologies, such as eye-tracking and EEG analysis, is also contributing to the widespread adoption of drowsiness detection systems. These technologies offer more accurate and reliable results, making drowsiness detection systems increasingly popular and effective in promoting road safety.

Chapter-4

Literature Review

According to Riad Alharbey, Mohamed M. Dessouky, Ahmed Sedik, Ali I. Siam And Mohamed A. Elaskily et al.[1], here the author examines two approaches to detect fatigue in high-performance car drivers: a machine learning approach and a deep learning approach. The aim is to create an effective fatigue detection system. The machine learning approach focuses on analyzing EEG (electroencephalogram) signals to predict driver behavior and identify fatigue. The proposed algorithm employs several machine learning classifiers, including SVM (**support vector machine**), RF (**Random forest**), LR (**logistic regression**), MLP, KNN (k-nearest neighbor), and QDA, to determine the best method for detecting fatigue. Results show that SVM provides the highest accuracy and shortest detection time, with 98% accuracy and a detection time under 0.18 seconds. On the other hand, the deep learning approach utilizes video streaming recordings of the driver to detect fatigue. The proposed algorithm combines CNN, ConvLSTM, and a hybrid CNN-ConvLSTM model, with the best results coming from the CNN model. A combination of CNN, ConvLSTM, and hybrid CNN-ConvLSTM models was used, and results showed that CNN was the best model, achieving 99% detection accuracy with a detection time of 10.61 seconds. This paper presents a thorough evaluation of machine learning and deep learning approaches for fatigue detection in high-performance car drivers. The results demonstrate the potential of these approaches to accurately and efficiently detect driver fatigue. The authors plan to test the two approaches under different driving conditions and investigate combining the two for better results.

They also plan to integrate the detection system with IOT systems and create a prototype for the proposed algorithm. Overall, the paper provides valuable contributions to the field of fatigue detection for high-performance car drivers. The use of EEG signals and video streaming recordings for fatigue detection has shown promising results and the authors' plans for future work will further improve the effectiveness of the system.

Future work will involve testing the two approaches with other datasets to determine their efficiency in various driving conditions. Efforts will also be made to combine the two approaches to improve results and integrate the system with IoT systems through cloud computing. Additionally, a prototype of the proposed algorithm will be developed.

According to V.Uma Maheswari, (Member, Ieee), Rajanikanth Aluvalu, (Senior Member,

Ieee),Mvv Prasad Kantipudi, (Senior Member, Ieee),Krishna Keerthi ChennamKetan Kotecha And Jatinderkumar R. Saini et al. [2], the "Driver Drowsiness Prediction Based on Multiple Factors Utilizing Image Processing Strategies" study explores the use of image processing techniques to identify drowsy drivers. The article presents a unified approach that combines the Eye and Mouth Closure Status (PERCLOS) and the calculated Facial Aspect Ratio (FAR) to assess driver fatigue. The system also incorporates textural gradient patterns and the detection of hand gestures to detect the driver's face, sunglasses, and hands covering their eyes or mouth. The study cites the rising number of road accidents due to driver fatigue as the driving force behind the research. Despite various drowsiness detection systems existing, the authors highlight that there is still a need for improvement in terms of efficiency, accuracy, cost, speed, and availability. The proposed method was tested on datasets such as NTHU-DDD, YawDD, and the newly proposed EMOCDS (Eye and Mouth Open Close Data Set) and demonstrated higher accuracy and efficacy in various circumstances.

The authors conclude that their approach offers a more comprehensive and accurate solution to the problem of driver drowsiness. The integration of multiple factors such as eye and mouth closure, hand gestures, and textural gradient patterns enhances the system's accuracy and performance. The authors suggest that their approach could be used in the transportation industry to alert drivers and prevent accidents caused by fatigue.

The research paper provides a valuable contribution to the field of driver drowsiness detection, demonstrating the effectiveness of image processing techniques in identifying drowsy drivers. The integration of multiple factors enhances the accuracy of the system, and the proposed approach holds promise for improving road safety and saving lives.

According to DUY-LINH NGUYEN , (Member, IEEE), MUHAMAD DWISNANTOPUTRO , (Member, IEEE), AND KANG-HYUN JO , (Senior Member, IEEE) et al. [3], this research paper presents a light-weight convolutional neural network architecture to recognize and categorize driver behaviors. The combination of feature extraction and classifier modules, along with the implementation of a convolution block attention mechanism, help the network to effectively extract features and minimize traffic collisions. The network has been trained and evaluated on three benchmark datasets, resulting in high classification accuracy. The problem of driver behavior is a growing concern as it is a main cause of traffic accidents, with distracted driving being defined as actions that take a driver's attention away from the road. The authors highlight the importance of developing systems to warn drivers and propose their network as a solution. The network's design balances accuracy and optimization, making it

cost-effective for deployment in modern or older vehicles. The authors believe their proposed light-weight CNN architecture can provide an effective and affordable solution for driver behavior recognition. The results of their study show promising accuracy rates, and the video tests demonstrate the effectiveness of the system. By providing accurate information and warning drivers in advance, this research aims to minimize traffic collisions and improve road safety.

Distracted driving is a significant contributor to traffic accidents, accounting for 8.7% of all traffic accident deaths in the United States in 2019. Distracted driving is defined as behaviors that interrupt a driver's focus, including cell phone use, eating, drinking, and adjusting the entertainment system. The study of distracted driving has been divided into three groups: visual distraction, manual distraction, and cognitive distraction. Visual distraction focuses on eye status and head posture analysis, while manual distraction tracks hand and foot activities on the gas and brake pedal. Cognitive distraction predicts the psychophysiological state of drivers, such as heart rate, blood pressure, and body temperature. Despite efforts by automakers to integrate analytical and driver warning devices in vehicles, most are still in the testing phase and are costly to deploy in older cars.

According to Seyed Kian Mousavikia, Erfan Gholizadehazari, Morteza Mousazadeh, and Siddika Berna Ors Yalcin et al. [4], this paper presents the design and implementation of a driver drowsiness detection system using a modified RiscV processor on a field-programmable gate array (FPGA). The system utilizes a Convolutional Neural Network (CNN) trained to classify four primary driver's expressions, including distraction, natural, sleep, and yawn, with an accuracy of 81.07% on validation data. To overcome FPGA memory limitations, the authors optimized the written C code for the trained CNN through the use of dynamic fixed-point data types and dynamic memory allocations. Additionally, the RiscV processor was modified by adding three custom instructions, custom store, conv2d (2×2), and multiply and accumulation (MAC), to enhance the computation rate.

The modified processor with these custom instructions showed the best result in terms of latency, with an improvement factor of 1.7 over the base processor and 1.25 over the processor with only custom store and MAC.

The authors highlight the importance of driver drowsiness detection as it is a leading cause of human casualties in traffic accidents, along with property damage. The use of neural networks, particularly CNNs, in drowsiness detection is widely accepted due to their high accuracy in classification. The implementation of the trained CNN model on the FPGA was chosen due to

the flexibility it offers in changing the hardware architecture. The authors also considered the open-source RiscV as the instruction set architecture for the embedded processor for its simplicity and ease of modification.

In conclusion, this paper provides a comprehensive approach to designing a driver drowsiness detection system using a modified RiscV processor on an FPGA. The optimized CNN and custom instructions added to the processor showed improvement in terms of latency. The authors aim to minimize traffic accidents through the detection of drowsiness and distraction, and the design of the system is compatible with future ASIC implementation.

According to Hiroaki Hayashi , Mitsuhiro Kamezaki (Member, IEEE), And Shigeki Sugano (Fellow, IEEE) et al. [5], in this research paper, the authors present a system designed to detect and address health-related accidents while driving. The system employs computer vision to observe the driver's body posture and eye movements. A set of 1600 images of posture and 2800 images of eyes were gathered and annotated, with 70% used to train a k-NN and CNN classifier, and the remaining 30% reserved for testing purposes. The posture classification accuracy was found to be between 91-100%, and the accuracy of eye state classification was 99-100%. In a follow-up experiment, the system was tested in typical driving scenarios and produced the expected results. The mean and deviation of each signal were calculated and a coefficient k was used to calculate the driver's Comfort Level (CL). If the duration of a signal was outside of normal limits, the CL would exceed 50, triggering verbal interaction with the driver. Finally, the system was tested to determine its ability to detect fake headaches. Five participants were instructed to pretend to have a headache, and the system was able to accurately identify all five instances. Additionally, the system was able to differentiate normal behavior and did not initiate verbal interaction. The final CL was based on the maximum CL of each signal, and if the final CL was greater than 50, verbal interaction was initiated. The results of this experiment demonstrate that the system can effectively detect and address health-related accidents while driving. In conclusion, the authors present a compelling solution for improving road safety by detecting and addressing health-related accidents. The results of the experiments indicate that the system can accurately identify signals and initiate verbal interaction to prevent accidents. The system can be further improved by increasing the training data and refining the model for improved accuracy.

According to Ayman Altameem , Ankit Kumar , Ramesh Chandra Poonia, Sandeep Kumar And Abdul Khader Jilani Saudagar et al. [6], Driver drowsiness is a major cause of car accidents

and has severe implications for roadsafety. To avoid fatal accidents, various drowsiness detection technologies have been developed to monitor a driver's inattention and alert the driver. These technologies use sensors in self-driving cars to constantly monitor the driver's facial expressions and detectfacial landmarks to extract the driver's state of expression and determine their driving safety. The proposed system will be integrated with the vehicle's electronics to track thevehicle's statistics and provide more accurate results.

The authors have implemented real-time image segmentation and drowsiness detection using machine learning methodologies. The algorithm uses a Support Vector Machines (SVM) based emotion detection methodto detect facial expressions. The algorithm was tested under different luminance conditions and outperformed current research with 83.25% accuracy in detecting facial expression changes. The authors aim to develop a robust and practical drowsiness recognition system to reduce the number of accidents caused by driver drowsiness. The U.S. Department of Transportation is working towards developing smart vehicles to avoid such accidents, and this research provides an improved and precise technique for identifying drowsiness. Despite promising progress in ongoing research, some core issues still need to be addressed. Currently, detection methods use the driver's behavior or physiological changes and the vehicle's reactions to the driver's behavior to detect drowsiness, but each method has its own advantages and limitations.

According Burcu Kir Savaş And Yaşar Becerikli et al. [7], A novel approach to detecting driver fatigue in real-time is presented in this article with the introduction of a Multi-tasking Convolutional Neural Network (ConNN) model. Thismodel leverages both eye and mouth characteristics to monitor changes and identify drowsiness in drivers. The Multi-task ConNN stands out from prior research by incorporating both mouth and eye information into a single model, as opposed to treatingthem as separate entities. The level of fatigue is measured by determining the duration ofeye closure (PERCLOS) and the frequency of yawning (FOM). The proposed model wastested on the YawdDD and NthuDDD datasets, resulting in an impressive accuracy rate of 98.81% for detecting driver fatigue. The significance of this research is highlighted bystatistics from organizations such as the World Health Organization and the American Automobile Association, which highlight driver fatigue as a leading cause of traffic accidents. Past studies have utilized a range of algorithms including radial basis functionneural network, artificial neural networks, support vector machine, AdaBoost, principal component analysis, and others. Nevertheless, the Multi-task CNN model presented inthis article represents a new level of accuracy in detecting driver fatigue by combining eye and mouth information

into a single classification model.

The model can be integrated into existing driver assistance systems or used as a standalone system to detect driver fatigue in real-time. The results of this study provide a significant contribution to the field of driver fatigue detection and highlight the potential of using deep learning techniques for this application. The proposed model has the potential to be used in vehicles to improve road safety, prevent traffic accidents, and save lives. Past studies have utilized a range of algorithms including radial basic function neural network, artificial neural networks, support vector machine, AdaBoost, principal component analysis, and others.

The proposed model was tested on the YawdDD and NthuDDD datasets, resulting in an impressive accuracy rate of 98.81% for detecting driver fatigue. The significance of this research is highlighted by statistics from organizations such as the World Health Organization and the American Automobile Association, which highlight driver fatigue as a leading cause of traffic accidents.

According to Mika Sunagawa, Shin-ichi Shikii, Wataru Nakai, Makoto Mochizuki, Koichi Kusukame, and Hiroki Kitajima et al. [8], the paper introduces a new model for detecting drowsiness in drivers, which can identify different levels of drowsiness, ranging from mild to severe. The model is based on the idea that posture-related indicators can reveal weaker forms of drowsiness that drivers might not be aware of. To test this hypothesis, the authors conducted experiments on 50 drivers in simulated driving conditions and measured the effectiveness of posture and conventional indicators in detecting drowsiness at different stages. The results showed that posture information was a valuable tool in detecting weak drowsiness and, when combined with blink information, the model was able to accurately detect all levels of drowsiness.

The evaluation of the model yielded an F1-score of 53.6% and a root mean square error of 0.620, demonstrating its effectiveness. One of the unique features of this model is that it only requires information from the driver's blink and posture, which can be obtained without any restrictive equipment. The authors believe that this model has the potential to be used in developing warning systems for drowsy drivers and in systems that can prevent drowsiness before it occurs.

The authors found that posture information was indeed a useful indicator for weak drowsiness detection, and combined it with blink information to create a comprehensive model. The results of the evaluation showed that the combined model was successful in detecting all stages of drowsiness, with an F1-score of 53.6% and a root mean square error of 0.620. The model is

based on the assumption that posture-related indices can indicate weak drowsiness that drivers might not be aware of. The authors first evaluated the sensitivity of posture and conventional indices for different stages of drowsiness and then designed a model that combined these indices to detect all levels of drowsiness.

The paper presents a drowsiness detection model that covers the entire range of drowsiness, from weak to strong. The model combines several indices sensitive to weak and strong drowsiness and uses posture information and blink information to detect drowsiness. The information required for the model can be measured without restrictive equipment, making it practical for use in driver drowsiness warning systems.

According to Belal Alshaqai; Abdullah Salem Baquhaizel; Mohamed El Amine Ouis; Meriem Boumehed; Abdelaziz Ouamri; Mokhtar Keche et al. [9], in this paper they summarized that how driver drowsiness detection system helps in aiding the prevention of accidents passenger and commercial vehicles. The study "Driver Drowsiness Detection System" focuses on developing an innovative module for Advanced Driver Assistance System (ADAS) aimed at mitigating road accidents caused by driver drowsiness. The module utilizes artificial intelligence algorithms to automatically detect signs of drowsiness by analyzing visual information and monitoring the driver's face and eyes. The proposed algorithm calculates the Percentage of Eye Closure (PERCLOS), a scientifically-supported indicator of drowsiness. The authors acknowledge the significant impact of driver fatigue and drowsiness on road accidents and highlight the need for innovative technologies to monitor drivers and assess their level of attention while driving. They conduct a comprehensive review of related works in the field of drowsiness detection systems that use vision-based techniques, including studies by Malla et al. (Haar algorithm), Vitabile et al. (infrared camera), Bhowmick and Kumar (Otsu thresholding), Hong et al. (optimized Jones and Viola method), and Tian and Qin (YCbCr color space). These studies focus on identifying and tracking the driver's face and eyes and evaluating eye states using various algorithms and techniques. The paper presents a valuable contribution to the field by proposing a new algorithm for measuring PERCLOS, which has the potential to enhance road safety and reduce the number of accidents caused by driver drowsiness.

In conclusion, the paper provides a comprehensive review of the current state-of-the-art in driver drowsiness detection systems and demonstrates the authors' efforts to improve road safety through innovative technology solutions. The proposed algorithm for measuring PERCLOS holds great promise for reducing the number of road accidents caused by driver fatigue and drowsiness.

According to Tianyi Hong, Huabiao Qin et al.[10],they have presented a new approach to detecting driver drowsiness using an embedded system in their paper titled "Drivers Drowsiness Detection in Embedded System". The current drowsiness detection methods have limitations in real-time detection and accuracy when the driver's head is tilted. To overcome these challenges, the authors have proposed a four-step method that uses image processing techniques. The method includes face detection, eye location initialization, eye tracking, and eye state identification. The method uses a combination of Haar-like features and object tracking to detect and track the eyes, and eye state is determined through PERCLOS. The experimental results show that the proposed method is efficient and accurate in real-time detection of drowsiness. However, further evaluations under different lighting conditions and with different subjects are suggested to better understand its performance.

The proposed method consists of four stages; face detection, eye location initialization, eye tracking, and eye state identification. The authors use a feature-based face detection method using Haar-like features and an object tracking method to keep track of the eyes. The eye state is then identified using PERCLOS. The authors claim that the proposed method is more efficient and accurate compared to traditional methods and results in real-time detection.

In conclusion, the authors have proposed a novel approach to detecting drowsiness in drivers using an embedded system. The proposed method addresses the problems of real-time detection and head tilt, which are prevalent in traditional drowsiness detection systems. The experimental results show the effectiveness of the proposed method, and further research could be conducted to evaluate its performance under different conditions.

Aims and Objectives of the Study

The main aim of a driver drowsiness detection system is to identify when a driver becomes drowsy and alert them to take a break or pullover, in order to prevent drowsy driving accidents.

The objective of a study on driver drowsiness detection systems may be to:

- (i)** Evaluate the accuracy and effectiveness of existing drowsiness detection technology and algorithms.
- (ii)** Develop and improve upon new and existing drowsiness detection systems, including the integration of multiple sensor modalities for more accurate detection.
- (iii)** Test the real-world effectiveness of drowsiness detection systems in various driving scenarios and conditions.
- (iv)** Study the individual differences in drowsiness and their impact on the effectiveness of drowsiness detection systems.
- (v)** Investigate ethical and privacy concerns related to the use of drowsiness detection systems in vehicles.

Chapter-6

Research Gaps

While reviewing the project entitled “Driver Drowsiness Detection System” I found some potential research gaps in this field all are listed below:

- (i) **Lack of real-world testing:** Most drowsiness detection systems have only been tested in controlled laboratory environments, leading to a gap in understanding their effectiveness in real-world driving conditions.
- (ii) **Limited sensor modalities:** Currently, most drowsiness detection systems rely on a single modality, such as eye tracking, which may not be sufficient to accurately detect drowsiness in all scenarios.
- (iii) **Interference from external factors:** Factors such as sunglasses, glare from the sun, or even changes in lighting conditions can interfere with the accuracy of drowsiness detection systems.
- (iv) **Individual differences:** Drowsiness affects individuals differently, making it challenging to develop a one-size-fits-all drowsiness detection system.
- (v) **Limited generalization:** Drowsiness detection systems trained on one set of data may not generalize well to other populations or driving conditions.
- (vi) **Ethical considerations:** The use of drowsiness detection systems raises ethical questions about privacy, data security, and the potential misuse of the technology.
- (vii) **Integration with vehicle systems:** Developing a seamless integration of drowsiness detection systems with the vehicle's other systems, such as the infotainment system, is a major challenge.

The scope of a study on driver drowsiness detection systems may include the following aspects:

- (i) **Literature review:** An extensive review of previous studies and research on drowsiness detection systems, including a review of existing drowsiness detection algorithms and technologies.
- (ii) **Methodology:** A description of the methodology used to evaluate the performance of drowsiness detection systems, including the experimental setup, data collection methods, and statistical analysis techniques.
- (iii) **Experimentation:** The experimental evaluation of existing drowsiness detection systems and the development and testing of new drowsiness detection algorithms and technologies.
- (iv) **Results and analysis:** The presentation and analysis of the experimental results, including the accuracy and reliability of the drowsiness detection systems in detecting drowsiness.
- (v) **Comparison:** A comparison of the performance of different drowsiness detection systems, including an analysis of the strengths and weaknesses of each system.
- (vi) **Discussion:** A discussion of the implications of the results for the development of future drowsiness detection systems, including recommendations for future research.

8.1 Flow Chart

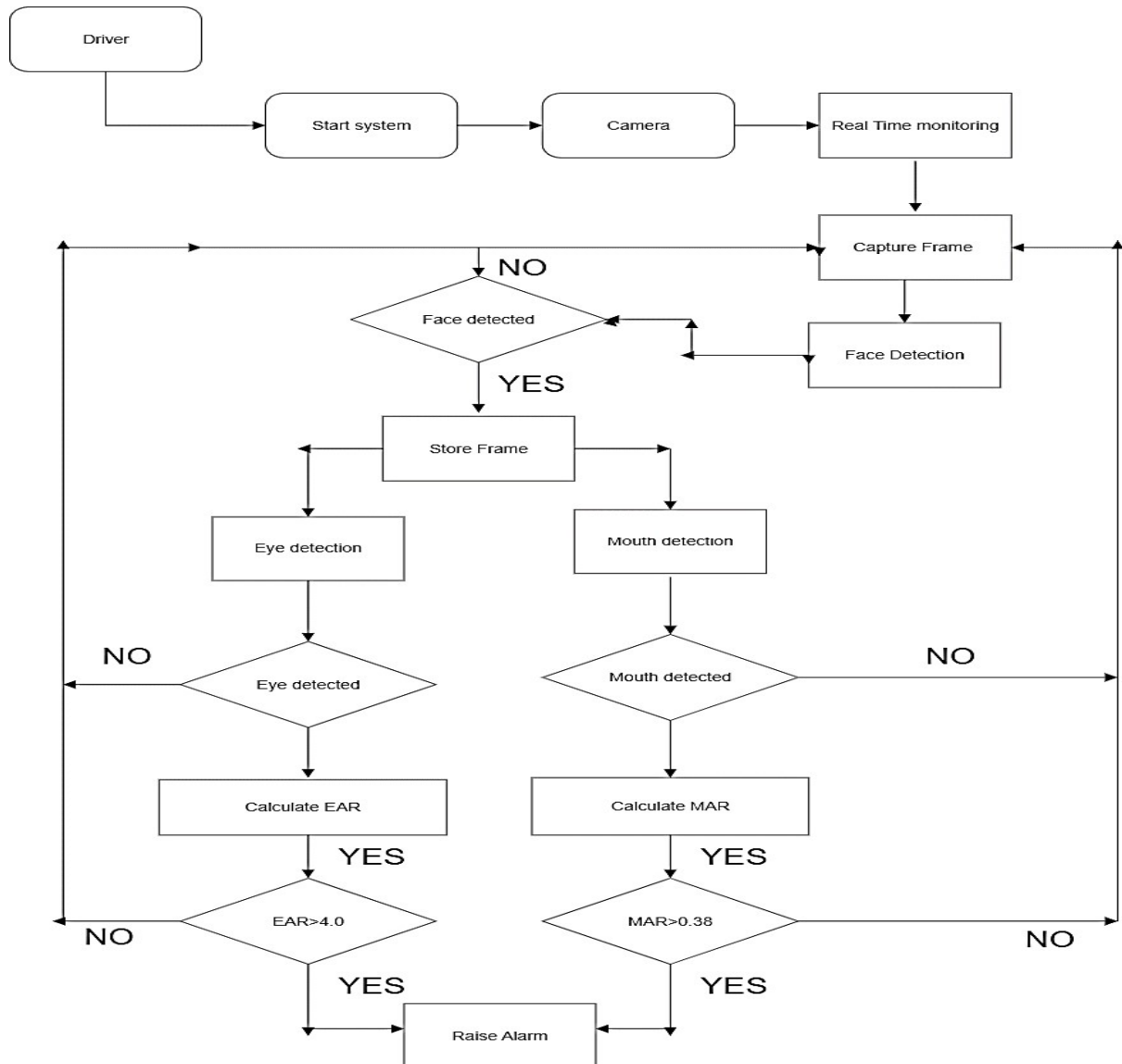


Figure 8.1.1: Flow chart showing the entire process of Drowsiness Detection System

The proposed technique is primarily based on eye blinking and yawning of drivers which can be behavioral measures. The aim of this project is to detect closed eyes or opened mouth, this is yawning and to alert the driver. This is done by placing a camera or recording device in front of the driver and capturing real time video continuously using OpenCV and DLib. The application is executed in python and processing is done in the laptop's camera. The work is mostly divided

into three steps: (i) Face, eye and mouth detection. (ii) Eye closure detection. (iii) Open Mouth detection.

- (i) **Face, eye and mouth detection:** In this step face is detected by the use of DLib library. Shape predictor that is implemented in DLib library is used to find facial landmarks. The Predictor gives 68 landmark points that can be applied to localize regions of the face such as eyes, eyebrows, nose, ears and mouth. Figure 1 gives the facial landmarks that can be detected via DLib. Therefore, by applying facial landmarks detection eyes and mouth can be localized and detected.
- (ii) **Eye Closure Detection:** Each eye is characterized by 6 coordinates as in figure 2.2. An equation called Eye Aspect Ratio (EAR) which reflects the relation between width and height of coordinators can be derived. The distance between vertical eye landmarks is computed in numerator and those of horizontal eye landmarks are calculated in the denominator using the formula for Euclidean distance.

$$\text{Eye Aspect Ratio} = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_4\|}$$



Figure 8.1.2: Facial Recognition Associated With Eyes

- (iii) **Open Mouth Detection:** Yawning is characterized by wide opening of mouth. Facial landmarks can be used to detect an opening, mouth. Mouth is characterized with the help of 20 coordinates as shown in figure 2.3. Using these coordinates, the distance between the lips is calculated which is the difference between the top lip and bottom lip and this lip distance is used to determine whether the driver's mouth is open. If the lip distance is greater than a threshold amount the subject is determined to be yawning and an alert is given accordingly.

$$\text{Mouth Aspect Ratio} = \frac{\|p_2 - p_8\| + \|p_3 - p_7\| + \|p_4 - p_8\|}{2\|p_1 - p_5\|}$$

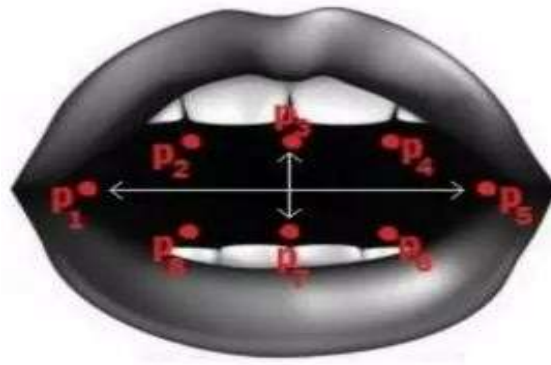


Figure 8.1.3: Facial Recognition Associated With Mouth

8.2. Description of tools used in this project:

- (i) **OpenCV:** OpenCV (Open-source Computer Vision) is the Swiss Army knife of computer vision. It has a wide range of modules that can help us with a lot of computer vision problems. But perhaps the most useful part of OpenCV is its architecture and memory management. It provides you with a framework in which you can work with images and video in any way you want, using OpenCV's algorithms or your own, without worrying about allocating and reallocating memory for your images. Open CV libraries and functions are highly optimized and can be used for real time image and video processing. OPENCV's highly optimized image processing function are used by the author for real time image processing of live video feed from camera.
- (ii) **DLib:** DLib is a modern C++ toolkit containing machine learning algorithms and tools for creating complex software in C++ to solve real world problems. It is used in both industry and academia in a wide range of domains including robotics, embedded devices, mobile phones, and large high performance computing environments. DLib's open-source licensing allows you to use it in any application, free of charge. Open-Source DLib library is used by the author for implementation of CNN (Neural Networks). Highly optimized pre-learned facial shape predictor and detector functions are used by the author for detection of facial landmarks. Facial landmarks were further used for extracting eye coordinates.
- (iii) **Python:** Python is an object-oriented programming language created by Guido Rossum in 1989. It is ideally designed for rapid prototyping of complex applications. It has interfaces to many Operating system calls and libraries and is extensible to C or C++. Many large companies use the Python programming language including NASA, Google, YouTube, BitTorrent, etc. Python

is widely used in Artificial Intelligence, Natural Language Generation, Neural Networks and other advanced fields of Computer Science. Python had deep focus on code readability. Python language is used by the author due to his cross-platform compatibility as main coding language for algorithm. OpenCV and DLib libraries are integrated in python interpreter for using readymade optimized functions.

- (iv) **PyCharm:** Open-source community version of PyCharm platform is used as main coding editor.

8.3 Use Cases:

- (i) **Automotive industry:** Driver drowsiness detection systems can be integrated into vehicles to alert drivers, when they show signs of fatigue, helping to prevent accidents caused by drowsy driving.
- (ii) **Long-distance trucking:** Long-haul truck drivers who spend hours on the road can benefit from drowsiness detection systems, which can alert them when they need to take a break.
- (iii) **Public transportation:** Bus and train drivers can use drowsiness detection systems to stay alert during long shifts and to ensure the safety of passengers.
- (iv) **Military and aviation:** Pilots, military personnel, and other professionals who operate vehicles or machinery in high-stakes environments can use drowsiness detection systems to stay alert and focused.

8.4 Data Flow Diagram:

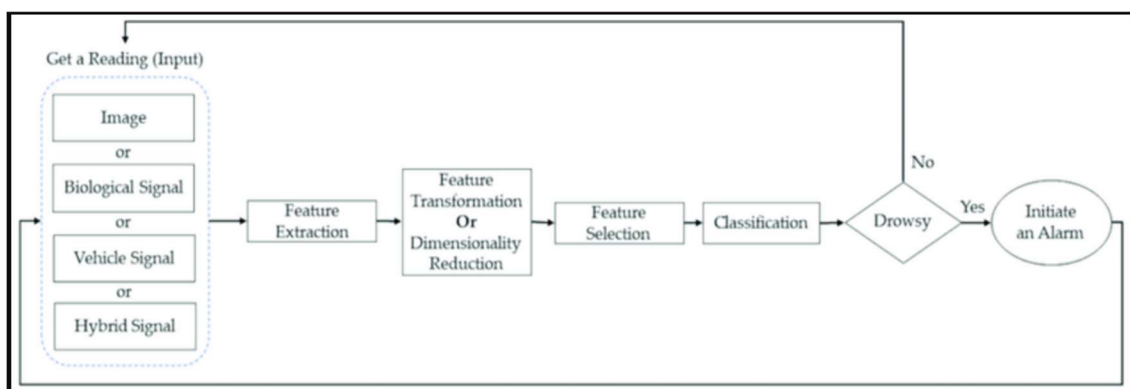


Figure 8.4.1: Data Flow Diagram

8.5 Entity Relationship:

The "entity relationship" of a driver drowsiness detection system refers to the relationships between different components or entities within the system. In the context of a drowsiness detection system, these entities could include:

- a) **The driver:** The person operating the vehicle and being monitored for signs of drowsiness.
- b) **The sensors:** The devices used to collect data on the driver's level of alertness, such as eye-tracking cameras, EEG sensors, or heart rate monitors.
- c) **The processor:** The component that processes the data collected by the sensors and determines the level of drowsiness of the driver.
- d) **The actuators:** The devices that trigger an event response, such as an alarm or a reduction in vehicle speed.

The entity relationship of a drowsiness detection system will typically involve the sensors collecting data on the driver, which is then processed by the processor to determine the level of drowsiness. Based on this determination, the actuators may be triggered to issue an alert or take other steps to prevent a potential accident. The relationships between these entities can be complex, and it is important to carefully design and test the system to ensure that it functions as intended.

8.6 Code:

```
Drowsiness_Detection.py X
C:\Users\hp>hp>Downloads> Drowsiness_Detection.py
1  from scipy.spatial import distance
2  from imutils import face_utils
3  from pygame import mixer
4  import imutils
5  import dlib
6  import cv2
7
8
9  mixer.init()
10 mixer.music.load("music.wav")
11
12 def eye_aspect_ratio(eye):
13     A = distance.euclidean(eye[1], eye[5])
14     B = distance.euclidean(eye[2], eye[4])
15     C = distance.euclidean(eye[0], eye[3])
16     ear = (A + B) / (2.0 * C)
17     return ear
18
19 thresh = 0.25
20 frame_check = 20
21 detect = dlib.get_frontal_face_detector()
22 predict = dlib.shape_predictor("models/shape_predictor_68_face_landmarks.dat")
23
24 (lStart, lEnd) = face_utils.FACIAL_LANDMARKS_68_IDXS["left_eye"]
25 (rStart, rEnd) = face_utils.FACIAL_LANDMARKS_68_IDXS["right_eye"]
26 cap=cv2.VideoCapture(0)
27 flag=0
28 while True:
29     ret, frame=cap.read()
30     frame = imutils.resize(frame, width=450)
31     gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
32     subjects = detect(gray, 0)
33     for subject in subjects:
34         shape = predict(gray, subject)
35         shape = face_utils.shape_to_np(shape)
36         leftEye = shape[lStart:lEnd]
37         rightEye = shape[rStart:rEnd]
38         leftEAR = eye_aspect_ratio(leftEye)
39         rightEAR = eye_aspect_ratio(rightEye)
40         ear = (leftEAR + rightEAR) / 2.0
41         leftEyeHull = cv2.convexHull(leftEye)
42         rightEyeHull = cv2.convexHull(rightEye)
43         cv2.drawContours(frame, [leftEyeHull], -1, (0, 255, 0), 1)
44         cv2.drawContours(frame, [rightEyeHull], -1, (0, 255, 0), 1)
45         if ear < thresh:
46             flag += 1
47             print (flag)
48             if flag >= frame_check:
49                 cv2.putText(frame, "*****ALERT!*****", (10, 30),
50                     cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 0, 255), 2)
51                 cv2.putText(frame, "*****ALERT!*****", (10,325),
52                     cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 0, 255), 2)
53                 mixer.music.play()
54             else:
55                 flag = 0
56         cv2.imshow("Frame", frame)
57         key = cv2.waitKey(1) & 0xFF
58         if key == ord("q"):
59             break
60     cv2.destroyAllWindows()
61 cap.release()
62
```

Figure 8.6.1: Implementation of Code

Chapter-9

Result and Discussion

The proposed method was initially implemented using a laptop with attached webcam of 15 frames per second. The system was tested. Fig.7.1 shows the output when the subject is yawning. The visual output contains eye and mouth aspect ratio and the audio output is an alert message which include the signal sleepy. Facial landmarks are detected and displayed as shown in fig 7.1.

In fig 7.1 the eyes of subject are open and corresponding EAR value, MAR value and status is displayed in turn an alert alarm was also generated as audio output.

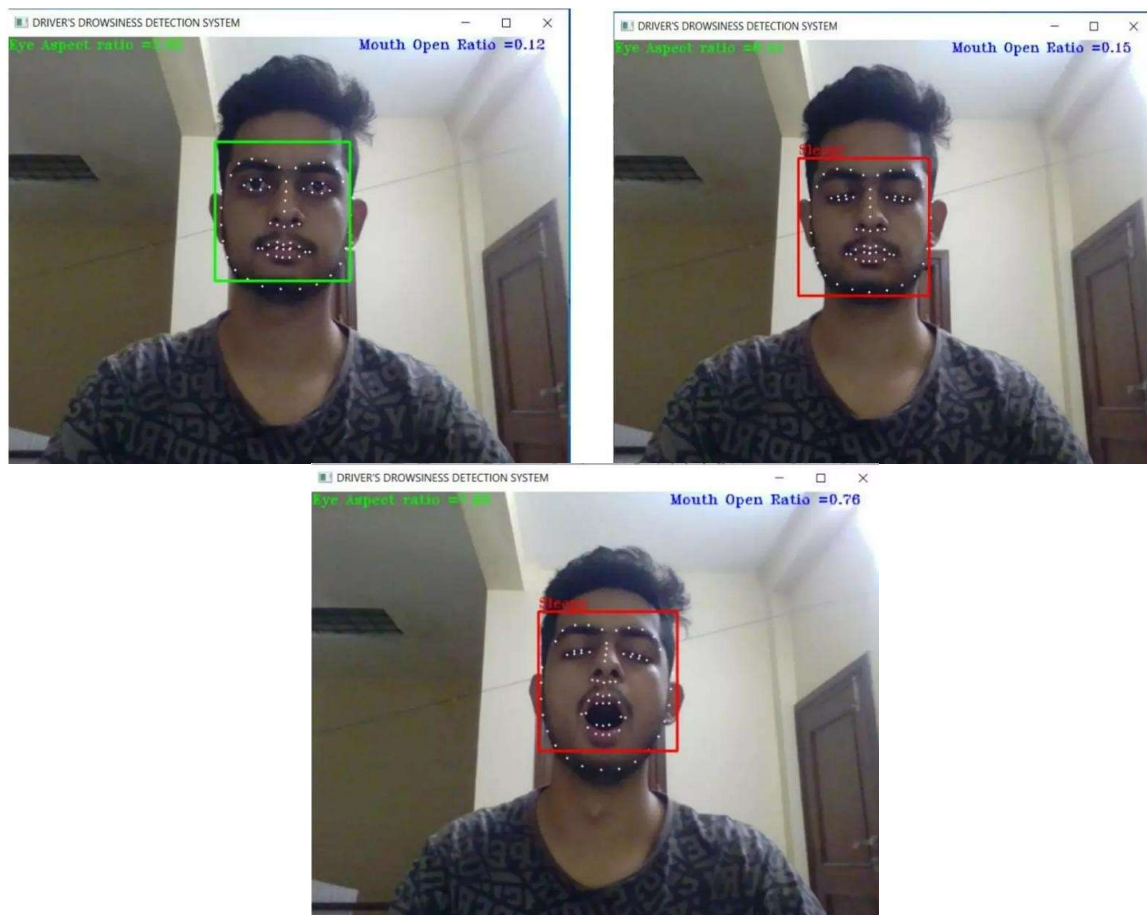


Figure 9.1: Result of Eyes and Mouth Aspect Ratio

Chapter-10

Conclusion

This project looks at how to detect tiredness in a driver in real time by looking at eyes closure and yawning. This technology has the advantage of detecting tiredness early on and sounding an alarm before an accident occurs. The use of OpenCV is considered to be more suitable for this application based on the design of the proposed work because it meets the relevant requirements such as cost, power, and size. Face, eye and mouth are easily detected by this technology, and these are captured using a webcam. The technology can detect whether the eyes and mouth were open or closed during monitoring. A warning signal will be issued if the eyes have been closed for an extended period of time or if yawning is detected.

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